Mohamed Emam

Biomechanics Researcher · AHPRA-Registered Physiotherapist

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PROFESSIONAL SUMMARY

Biomechanics researcher and AHPRA-registered physiotherapist completing a joint PhD at Semmelweis University (Budapest) and Cairo University (2022–2026). I bridge clinical rehabilitation with cutting-edge biomechanics using markerless motion capture (OpenCap) and statistical parametric mapping. Nine years of clinical and academic experience, 8 peer-reviewed publications in Q1/Q2 journals, and full general registration with AHPRA. Research spans gait analysis, neurorehabilitation, electrical stimulation, and aging physiology. Seeking postdoctoral fellowship opportunities.

EDUCATION

Joint PhD in Health Sciences 2022 – 2026
Semmelweis University, Budapest & Cairo University, Egypt
Postural control & proprioceptive rehabilitation in cervicogenic headache; Resistance training on the cost of walking & muscle architecture in healthy older adults. (surface EMG & machine learning analysis of biomechanical data)

MSc in Physical Therapy 2015 – 2019
Cairo University, Egypt

BSc in Physical Therapy (Honours) 2008 – 2013
Cairo University, Egypt

PROFESSIONAL EXPERIENCE

Assistant Lecturer 2016 – Present
Kafr El-Sheikh University, Egypt
Teaching musculoskeletal physiotherapy, lab demonstrations, curriculum development, student research mentorship.

Founder & Physiotherapist 2015 – Present
Start-Rehabilitation Centre, Kafr El-Sheikh
Clinical management of musculoskeletal & orthopaedic conditions; manual therapy and exercise rehabilitation; clinic operations.

R&D Team Lead 2022 – 2025
Physioway (Physiotherapy Technology Company)
Directed development of clinical applications (scoliosis assessment tool, vestibular rehabilitation app).

SELECTED PUBLICATIONS

1. **Exploring the Psychological Distress, Socioeconomic Hardship, and Academic Performance Among University Students During a Period of National Crisis.** *Social Sciences & Humanities Open*, 2026 (accepted). Q1
2. **Using Machine Learning-Based Classification of Postural Stability in Cervicogenic Headache Patients.** *Life*, 16(7):1061, 2026. Q1
3. **The Role of Hydrotherapy in Enhancing Recovery after Knee Arthroplasty: A Systematic Review and Meta-Analysis.** *Healthcare*, 14(13):2005, 2026. Q1

4. **Factors Influencing Health-Related Quality of Life in Children with Leukaemia in Saudi Arabia: A Cross-Sectional Study.** *BMC Pediatrics*, 2026. Q1
5. **The Effect of Surface EMG Biofeedback on Dysphagia in Stroke Patients: A Systematic Review.** *Developments in Health Sciences*, 2026 (accepted).
6. **Impact of Russian Current Therapy Targeting the Anterior Tibial Group on Balance and Fall Risk in Post-Stroke Patients: An RCT.** *Neurological Sciences*, 2026.
7. **Wrist Muscle Performance in Students with Chronic Nonspecific Neck Pain: An Isokinetic Assessment.** *BMC Musculoskeletal Disorders*, 2026.
8. **Effect of Gaze Direction Recognition Task on Pain, ROM and Functional Activities in Cervicogenic Headache Patients.** *BMC Neurology*, 2026.
9. **Proprioceptive Training Reduces Headache Burden and Centre of Pressure Path Length in Cervicogenic Headache: An RCT.** *Physiology International*, 2025.
0. **Effect of Different Types of Electrical Stimulation on Postural Control and Gait After Stroke: A Systematic Review and Meta-Analysis.** *Physiology International*, 112(4), 2025. Q2
1. **Blood Flow Restriction Exercise Protocols for Elderly Populations: A Scoping Review.** *Journal of Clinical Medicine*, 14(12):4185, 2025. Q1
2. **Proprioceptive Training Improves Postural Stability and Reduces Pain in Cervicogenic Headache Patients: An RCT.** *Journal of Clinical Medicine*, 13(22):6777, 2024. Q1
3. **Effect of Cervicogenic Headache on Myoelectrical Activities of Suboccipital Muscles, ROM and Functional Activities at Different Ages.** *Medical Journal of Cairo University*, 2019.

RESEARCH INTERESTS

- **Gait & movement biomechanics:** markerless motion capture (OpenCap), musculoskeletal modelling, statistical parametric mapping
- **Postural control & proprioceptive rehabilitation in cervicogenic headache:** surface EMG, posturography and machine-learning analysis of biomechanical data
- **Neurorehabilitation:** electrical stimulation, gait retraining and fall-risk reduction after stroke
- **Aging physiology:** eccentric vs. concentric resistance training, walking economy, muscle architecture

TECHNICAL SKILLS

- **Motion capture:** OpenCap (markerless), marker-based systems
- **Statistics:** SPM1D, Linear Mixed-Effects Models
- **Programming:** MATLAB, Python, R, SPSS
- **Imaging:** MRI muscle segmentation & analysis
- **Assessment:** Isokinetic dynamometry, surface EMG
- **Design:** RCTs, systematic reviews, meta-analyses

AWARDS, REGISTRATION & LANGUAGES

- **AHPRA** — Physiotherapist, Full General Registration (No. PHY0002905130)
- **Scholarships & grants:** Stipendium Hungaricum · Global Korea Scholarship · SE 250+ Excellence PhD (2025, 2026) · EKÖP Doctoral Excellence Grant (2026) · Pannonia Scholarship (2026) · Researchers Supporting Project Grant, PNU (2025) · ECSS and Motor Control conference travel grants (2025, 2026)
- **Recognition:** Best Project Award, Biomedical Data Summer School, Semmelweis University (2026); Best Poster Award, 1st International Health Science Conference, Szeged (2025); 10 peer reviews for 6 international journals
- **Conferences:** ECSS Lausanne 2026 (poster) & Rimini 2025 (poster) · International Motor Control Conference, Amsterdam 2025 (oral) · 3rd Biomechanics Symposium, Budapest 2025 (poster) · 1st International Health Science Conference, Szeged 2025 (poster) · Biomedical Data Summer School, Budapest 2026 · ORPHEUS, Budapest 2026
- **Languages:** Arabic (native) · English (C1–C2, PTE 81) · Hungarian (A1)